

AP Precalculus Unit 1

Polynomial and Rational Functions

Complete Guided Notes

Purpose

Fourteen topic packets with precise definitions, worked reasoning, representation laboratories, challenge tasks, error analysis, technology decisions, AP-style synthesis, and exit tickets.

ECHS Mathematics

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Strengthened 2026 classroom edition

Topic 1.1 - Change in Tandem

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A function is more than a formula: it is a coordinated assignment between quantities. Precise language prevents common errors such as confusing a positive output with an increasing function, treating a few displayed points as the entire graph, or ignoring a contextual domain.

Learning targets

- Describe how input and output values vary together, including sign and direction on intervals.
- Connect verbal, numerical, graphical, and symbolic representations of a function.
- Construct and critique contextual graphs with justified axes, units, anchors, domain, range, and phases.

Language and structure

Term	Meaning in this topic
function	An assignment in which each allowed input has exactly one output.
domain	The set of allowed input values.
range	The set of outputs actually produced.
image / preimage	An image is an output; a preimage is an input producing a specified output.
covariation	How output values change as input values change.

Launch - reason before calculating

A delivery drone stays above ground during its flight. It rises for four minutes, hovers, and then descends. Explain why the statements “height is positive” and “height is increasing” communicate different information.

Concept 1: Assignments, domain, range, and notation

Core idea

The notation $f(a) = b$ means input a is assigned output b . Solving $f(x) = b$ asks for one or more preimages. Several inputs may share one output, but one input cannot be assigned two outputs by a function.

Worked example - annotate the reasoning

Problem. The displayed pairs are $(-2, 7), (0, 1), (3, -8), (5, -14)$. Find $f(3)$, the preimage of 1, and the displayed domain and range.

Model reasoning. $f(3) = -8$; the displayed preimage of 1 is 0; domain $\{-2, 0, 3, 5\}$ and range $\{-14, -8, 1, 7\}$.

Check your understanding

Independent

No calculator

A relation contains $(1, 4), (2, 4), (3, 7), (3, 9)$. Is it a function of the first coordinate? Justify and give one repair.

Concept 2: Sign, direction, and interval language

Core idea

The sign of $f(x)$ compares an output with zero. Increasing or decreasing compares outputs at ordered inputs. A function may be positive and decreasing, negative and increasing, or cross zero while continuing in one direction.

Worked example - annotate the reasoning

Problem. For $P(t) = -(t - 4)^2 + 20$ on $0 \leq t \leq 8$, identify where P is increasing and decreasing and determine whether P is positive throughout.

Model reasoning. The graph rises to the vertex at $t = 4$, so it increases on $[0, 4]$ and decreases on $[4, 8]$. The endpoint values are 4 and the maximum is 20, so all outputs are positive.

Check your understanding

Independent

No calculator

A table gives $g(1) = -9, g(2) = -5, g(3) = -2, g(4) = -1$. Describe sign and direction without using the word positive.

Concept 3: Constructing a graph from a changing story

Core idea

A contextual graph requires the independent quantity on the horizontal axis, the dependent quantity and units on the vertical axis, a realistic domain, and supplied anchors. Words such as constant, faster, slowing, maximum, and unchanged control direction, steepness, and shape.

Worked example - annotate the reasoning

Problem. At 6 a.m. the temperature is 24°C . It rises quickly, then more slowly, reaches 39°C at 2 p.m., and falls to 31°C at 9 p.m. Describe a defensible graph.

Model reasoning. Use time horizontally and temperature vertically. Plot $(6, 24)$, $(14, 39)$, $(21, 31)$. The first portion rises with decreasing steepness, reaches a maximum at 14, then decreases. Domain $6 \leq t \leq 21$.

Check your understanding

Independent

No calculator

A tank starts with 600 L, drains steadily for 10 minutes, pauses for 4 minutes, then drains more rapidly until empty. Sketch and label each phase.

Representation laboratory

Connect at least two representations

The table gives $q(0) = 12$, $q(2) = 18$, $q(5) = 18$, $q(8) = 7$. Write a verbal account consistent with the table, then state one feature the table does not determine between listed inputs.

Challenge laboratory

Challenge - justify every claim

Functions f and g satisfy $f(0) = g(0) = 2$, $f(2) = g(2) = 8$, and $f(5) = g(5) = 1$. A student claims they have identical covariation on $[0, 5]$. Decide whether this follows and describe two counterexample graphs.

Error clinic and calculator strategy

Common errors to diagnose

- Treating $f(x) > 0$ as evidence that f is increasing.
- Listing only displayed tick marks as the full domain of a continuing graph.
- Forgetting that repeated outputs are allowed.
- Drawing a contextual graph without units, anchors, or a defensible domain.

Technology decision

Technology can display and compare representations, but it cannot decide whether a contextual domain is sensible or whether interpolation is justified. Record the window and scale before citing a graph as evidence.

AP-style synthesis

Multi-part reasoning task

A solar system produces energy $E(t)$, in kilowatt-hours, during eight hours after sunrise.

t	0	2	4	6	8
$E(t)$	0	9	21	17	0

- A. State the displayed domain and range and interpret $E(4) = 21$.

- B. Describe sign and displayed direction without assuming unrecorded behavior.

- C. Sketch one continuous graph consistent with the data and label a possible maximum.

Exit ticket

Before you leave

1. Can two inputs have the same output in a function? Explain.

2. Give a function that is negative and increasing on an interval.

3. Name two required features of a contextual graph.

Topic 1.2 - Rates of Change

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Rates of change turn covariation into a measurable comparison. The same quotient can represent speed, price per item, temperature change per hour, or another context-specific quantity. A correct rate must be attached to its interval and interpreted with units.

Learning targets

- Calculate and interpret average rates of change from formulas, tables, graphs, and contexts.
- Compare rates on different intervals and estimate a rate at a point using nearby average rates.
- Simplify difference quotients and explain what the algebra reveals about changing rates.

Language and structure

Term	Meaning in this topic
average rate of change	$\frac{f(b)-f(a)}{b-a}$ over $[a, b]$, with output units per input unit.
secant line	A line through two graph points; its slope equals average rate of change.
difference quotient	An expression such as $\frac{f(a+h)-f(a)}{h}$, where $h \neq 0$.
local rate estimate	A short-interval secant rate used to estimate behavior near an input.
net change	The output difference $f(b) - f(a)$, before division by input change.

Launch - reason before calculating

A tank drops from 800 L to 620 L during the first 3 hours, then from 620 L to 500 L during the next hour. Compare the two average rates and explain why comparing only the two volume decreases is misleading.

Concept 1: Average rate, interval, and units

Core idea

For distinct inputs a, b , use $\frac{f(b)-f(a)}{b-a}$. Keep subtraction order consistent. The sign describes net direction, while magnitude describes output change per input unit. Contextual interpretation must name both quantities and units.

Worked example - annotate the reasoning

Problem. Customers are modeled by $C(t) = 120 + 18t - 2t^2$, where t is hours after opening. Find and interpret the average rate on $[1, 4]$.

Model reasoning. $C(1) = 136, C(4) = 160$, so the rate is $24/3 = 8$ customers per hour. The customer count increased by an average of 8 customers for each hour over the interval.

Check your understanding

Independent

No calculator

For $H(t) = 4t^2 - 3t + 10$, calculate the average rate on $[2, 5]$.

Concept 2: Tables, nonuniform inputs, and nearby intervals

Core idea

When input steps differ, divide each output change by its own input change. To estimate a rate at $x = c$, use a short interval near or around c and state that the result is an estimate. Very short intervals may magnify rounding or measurement error.

Worked example - annotate the reasoning

Problem. A sensor reports $P(3.8) = 41.2$ and $P(4.2) = 42.0$. Estimate the rate at $x = 4$.

Model reasoning. The symmetric estimate is $(42.0 - 41.2)/(4.2 - 3.8) = 2$ output units per input unit.

Check your understanding

Independent

No calculator

Given $Q(1.9) = 12.44$ and $Q(2.1) = 13.08$, estimate the rate at $x = 2$.

Concept 3: Symbolic difference quotients

Core idea

Expand $f(a+h)$ carefully, subtract the whole expression $f(a)$, factor out h , and only then cancel under the condition $h \neq 0$. The simplified expression shows how an interval rate depends on the starting input and interval length.

Worked example - annotate the reasoning

Problem. For $f(x) = x^2 - 5x + 2$, simplify $\frac{f(a+h)-f(a)}{h}$.

Model reasoning. The numerator is $2ah + h^2 - 5h = h(2a + h - 5)$, so the quotient is $2a + h - 5$.

Check your understanding

Independent

No calculator

For $g(x) = 3x^2 + 2x - 1$, simplify $\frac{g(a+h)-g(a)}{h}$.

Representation laboratory

Connect at least two representations

A graph passes through $(0, 4)$, $(2, 10)$, $(5, 7)$, $(9, 19)$. Compute the consecutive displayed average rates, describe one possible graph, and state what the four points do not determine.

Challenge laboratory

Challenge - justify every claim

Construct $f(x) = ax^2 + bx + c$ with $f(0) = 1$, average rate 5 on $[0, 2]$, and average rate 9 on $[2, 4]$. Verify all conditions.

Error clinic and calculator strategy

Common errors to diagnose

- Using net change as the rate without dividing by input change.
- Using a common denominator for table intervals of different lengths.
- Reporting a rate without units or interval.
- Claiming an exact point rate from a long secant interval.

Technology decision

Technology can evaluate models and compare short intervals efficiently. Keep enough precision, record the interval, and distinguish a decimal estimate from an exact conclusion. A calculator does not supply the contextual interpretation.

AP-style synthesis

Multi-part reasoning task

Water depth $D(t)$, in centimeters, is recorded at selected times.

t (min)	0	3	7	8	12
$D(t)$	18	27	35	33	21

- A. Calculate and interpret the average rate on $[0, 7]$.

- B. Compare the rates on $[3, 7]$ and $[7, 8]$ and interpret their signs.

- C. Explain why the table does not determine the exact time of greatest depth.

Exit ticket

Before you leave

1. State the average-rate formula and units.

2. Why must $h \neq 0$ in a difference quotient?

3. Why is a nearby secant rate usually called an estimate at a point?

Topic 1.3 - Rates of Change in Linear and Quadratic Functions

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Linear functions have constant rate. Quadratic functions do not, but their rates change in a constant pattern over equal input steps. This distinction supports model identification, construction, prediction, and comparison without relying only on graph appearance.

Learning targets

- Use first and second differences correctly, including the equal-spacing requirement.
- Construct linear or quadratic models from values, rates, and structural conditions.
- Derive and interpret how average rates of a quadratic change across equal intervals.

Language and structure

Term	Meaning in this topic
first difference	The change between consecutive outputs for equal input steps.
second difference	The change between consecutive first differences.
linear signature	Constant first differences for equally spaced inputs.
quadratic signature	Constant nonzero second differences for equally spaced inputs.
step size	The common input increment h used in finite differences.

Launch - reason before calculating

For $x = 0, 2, 4, 6$, outputs are 5, 13, 29, 53. A student says the data are not quadratic because first differences are not constant. Evaluate the claim.

Concept 1: Finite-difference signatures

Core idea

For equally spaced inputs, a nonconstant function with constant first differences is linear; constant nonzero second differences indicate a quadratic. If spacing is unequal, compare average rates instead of raw differences.

Worked example - annotate the reasoning

Problem. For $x = 0, 1, 2, 3, 4$, outputs are 3, 10, 21, 36, 55. Classify and find a rule.

Model reasoning. First differences 7, 11, 15, 19 have second difference 4, so $a = 2$. Since $f(0) = 3$ and $f(1) = 10$, $b = 5$. Thus $f = 2x^2 + 5x + 3$.

Check your understanding

Independent

No calculator

For outputs 1, 6, 17, 34, 57 at $x = 0, 1, 2, 3, 4$, find a rule.

Concept 2: Constructing a quadratic from values

Core idea

Three noncollinear points with distinct inputs determine a quadratic. When one input is 0, the constant term is immediate. Finite differences reveal the leading coefficient quickly; substitution then gives the remaining term.

Worked example - annotate the reasoning

Problem. A quadratic has $q(0) = 4$, $q(1) = 9$, $q(2) = 18$. Determine $q(x)$.

Model reasoning. First differences 5, 9 give second difference 4, so $a = 2$. Then $c = 4$ and $2 + b + 4 = 9$, so $b = 3$. Hence $q = 2x^2 + 3x + 4$.

Check your understanding

Independent

No calculator

Find the quadratic through $(0, 2)$, $(1, 7)$, $(2, 16)$.

Concept 3: How quadratic rates change

Core idea

For $q(x) = ax^2 + bx + c$, the average rate on $[x, x + h]$ is $a(2x + h) + b$. For fixed h , this is linear in the starting input. Consecutive rates separated by step h differ by $2ah$, and output second differences equal $2ah^2$.

Worked example - annotate the reasoning

Problem. For $f(x) = 2x^2 - 3x + 1$, compare rates on $[0, 2]$ and $[2, 4]$.

Model reasoning. Using $a(2x + h) + b$, rates are 1 and 9. They differ by $8 = 2ah$ when the starting input moves by 2.

Check your understanding

Independent

No calculator

For $q(x) = -x^2 + 6x$, find rates on $[0, 1]$, $[1, 2]$, $[2, 3]$.

Representation laboratory

Connect at least two representations

A table has inputs 0, 1, 3, 6 and outputs 2, 5, 17, 62. Explain why raw differences cannot prove a quadratic pattern. Compute consecutive average rates.

Challenge laboratory

Challenge - justify every claim

For $f(x) = ax^2 + bx + c$, prove $f(x+h) - 2f(x) + f(x-h) = 2ah^2$. Explain why b, c disappear.

Error clinic and calculator strategy

Common errors to diagnose

- Calling any changing first-difference table quadratic without checking second differences.
- Using raw differences when input increments are unequal.
- Assuming a quadratic pattern must continue outside the data domain.
- Equating a with the second difference when step size is not 1.

Technology decision

Regression helps with noisy data, but exact finite differences reveal structure more transparently. Inspect residuals, report spacing, and do not treat a high fit statistic as permission for unlimited extrapolation.

AP-style synthesis**Multi-part reasoning task**

For equally spaced inputs, P has values

x	0	1	2	3	4
$P(x)$	12	18	28	42	60

A. Justify a quadratic model using differences.

B. Determine $P(x)$.

C. Find the average rate on $[1, 4]$ and connect it to one-unit rates.

Exit ticket**Before you leave**

1. What spacing condition is required for raw finite differences?

2. For step 1, how is second difference related to a ?

3. Can a genuine quadratic have constant rate on all intervals?

Topic 1.4 - Polynomial Functions and Rates of Change

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Higher-degree polynomials model changing rates more flexibly than linear or quadratic functions. Their formulas, tables, and graphs must be read together: finite differences suggest degree, rates describe intervals, technology locates features, and context restricts where a model is meaningful.

Learning targets

- Use finite differences and polynomial structure to identify or justify a likely degree.
- Analyze polynomial rates, turning behavior, zeros, and representations without overclaiming from secant evidence.
- Use technology responsibly to locate features and critique a polynomial model's domain.

Language and structure

Term	Meaning in this topic
polynomial degree	The greatest exponent with nonzero coefficient.
finite-difference order	The level at which equal-step differences become constant.
turning point	A location where a graph changes between increasing and decreasing.
local rate estimate	A short-interval secant rate near an input.
model domain	The interval on which a formula is intended to represent a context.

Launch - reason before calculating

For $x = 0, 1, 2, 3, 4$, outputs are 1, 2, 9, 28, 65. Build successive difference rows, identify a likely degree, and explain why a contextual model need not remain valid for all real inputs.

Concept 1: Degree and finite differences

Core idea

For exact polynomial values at equally spaced inputs, a degree- n polynomial has constant n th differences and zero $(n + 1)$ st differences. This gives the least likely degree for exact data; measured data usually show only approximate patterns.

Worked example - annotate the reasoning

Problem. For $x = 0, 1, 2, 3, 4$, values are 2, 0, -2, 2, 18. Find the least polynomial degree.

Model reasoning. First differences -2, -2, 4, 16; second 0, 6, 12; third 6, 6. Least degree 3.

Check your understanding

Independent

No calculator

For outputs 3, 5, 19, 57, 131 at unit-spaced inputs, determine the least degree.

Concept 2: Rates, direction, and turning behavior

Core idea

Adjacent average rates show net increase or decrease. Changing rate signs can suggest turning behavior, but exact turning inputs may lie inside intervals. Equal endpoint values give average rate zero but do not imply constancy. A degree- n polynomial has at most $n - 1$ turning points.

Worked example - annotate the reasoning

Problem. For $P(x) = x^3 - 6x^2 + 9x + 2$, find one-unit rates from $x = 0$ through 4 and describe displayed direction.

Model reasoning. Values 2, 6, 4, 2, 6 give rates 4, -2, -2, 4: increase, then decrease, then increase, suggesting two turns.

Check your understanding

Independent

No calculator

For $q(x) = x^4 - 4x^2$, use values at -2, -1, 0, 1, 2 to describe direction.

Concept 3: Technology, local rates, and model limits

Core idea

A graphing utility can locate approximate zeros and extrema and estimate a local rate with a short interval. Evidence should include a window, domain, and precision. Polynomial end behavior can create impossible outputs outside a contextual interval.

Worked example - annotate the reasoning

Problem. For $F(x) = 0.5x^4 - 4x^2 + 3$, use $F(1.9) \approx -4.92395$ and $F(2.1) \approx -4.91595$ to estimate the rate near 2.

Model reasoning. The symmetric rate is $0.008/0.2 = 0.04$, close to zero, suggesting a nearly horizontal graph near 2 without proving an exact rate.

Check your understanding

Independent

Calculator permitted

A calculator gives $G(3.99) = 10.202$, $G(4.01) = 10.198$. Estimate the rate near 4.

Representation laboratory

Connect at least two representations

A polynomial has $f(-2) = 0$, $f(-1) = -6$, $f(0) = 0$, $f(1) = 6$, $f(2) = 0$. Describe signs and consecutive rates, then propose and verify a factored polynomial.

Challenge laboratory

Challenge - justify every claim

Construct a cubic with zeros $-2, 1, 4$ and $P(0) = 16$. Find its average rate on $[0, 2]$ and explain what the result does and does not communicate.

Error clinic and calculator strategy

Common errors to diagnose

- Applying finite differences to unequally spaced inputs.
- Claiming a degree- n polynomial must have exactly n real zeros or $n - 1$ turns.
- Treating an average rate of zero as proof of a constant function.
- Trusting a calculator graph without checking window and contextual domain.

Technology decision

Save the model equation, window, and relevant coordinates. Verify approximate features by substitution or sign checks. Residuals and context matter more than an impressive fit statistic, especially for high-degree extrapolation.

AP-style synthesis**Multi-part reasoning task**

A polynomial height model $H(t)$ is used on $0 \leq t \leq 10$. A graphing utility reports local extrema at approximately $(0.847, 30.319)$, $(4.665, 20.333)$, $(9.488, 38.973)$. Also $H(0) = 28$, $H(10) = 38$.

A. Describe increasing and decreasing intervals suggested by the extrema.

B. Determine absolute maximum and minimum on the contextual domain.

C. Explain why behavior for $t > 10$ should not automatically be interpreted as height.

Exit ticket**Before you leave**

1. Which finite differences are constant for exact cubic data?

2. Maximum turning points for degree 7?

3. Why can an algebraically valid polynomial be contextually invalid?

Topic 1.5 - Polynomial Functions and Complex Zeros

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A polynomial's real graph shows only real zeros, but its algebraic structure includes complex zeros as well. Counting multiplicity and using conjugate pairs makes it possible to construct, factor, and audit polynomials while explaining why a degree can exceed the number of visible intercepts.

Learning targets

- Apply the Fundamental Theorem of Algebra and conjugate-pair theorem to count zeros.
- Construct real-coefficient polynomials from real and complex zeros.
- Connect real-zero multiplicity to graph behavior while distinguishing complex zeros from intercepts.

Language and structure

Term	Meaning in this topic
Fundamental Theorem of Algebra	A degree- n polynomial has exactly n complex zeros counted with multiplicity.
conjugate pair	For real coefficients, a nonreal zero $a + bi$ requires $a - bi$.
multiplicity	The number of times a zero occurs as a factor.
real zero	A real input where the polynomial output is zero; an x -intercept.
irreducible quadratic	A real quadratic with negative discriminant and two nonreal conjugate zeros.

Launch - reason before calculating

A degree-5 polynomial with real coefficients has zeros 2, $-1 + i$, and $3i$. List every zero forced by the information and explain why no additional zeros are possible.

Concept 1: Counting zeros and conjugate structure

Core idea

Degree counts all real and nonreal zeros with multiplicity. If coefficients are real, nonreal zeros occur in conjugate pairs because conjugating the equation preserves the real coefficients. Consequently, an odd-degree real polynomial must have at least one real zero.

Worked example - annotate the reasoning

Problem. A degree-6 real polynomial has zeros 4, 4, and $1 - 2i$. What additional zeros are required, and how many zeros remain unspecified?

Model reasoning. The conjugate $1 + 2i$ is required. Counting multiplicity gives four known zeros, so two more zeros remain; they must be real or another conjugate pair.

Check your understanding

Independent

No calculator

A degree-7 real polynomial has zeros -3 (multiplicity 2), $2 + i$, and $5i$. How many zeros remain after including conjugates?

Concept 2: Constructing from complex zeros

Core idea

A conjugate pair $a \pm bi$ creates the real factor $(x - a)^2 + b^2$. Multiply these factors with real linear factors, then use a given point or leading coefficient to determine scale. Factored form often communicates zeros more clearly than expanded form.

Worked example - annotate the reasoning

Problem. Find the monic cubic with zeros 3 and $-2 \pm i$.

Model reasoning. The conjugate pair gives $(x + 2)^2 + 1 = x^2 + 4x + 5$. Thus $P = (x - 3)(x^2 + 4x + 5) = x^3 + x^2 - 7x - 15$.

Check your understanding

Independent

No calculator

Find the monic cubic with zeros 4 and $-1 \pm 2i$.

Concept 3: Multiplicity and the real graph

Core idea

Only real zeros create x -intercepts. At an odd-multiplicity real zero, the sign changes and the graph crosses; at even multiplicity, the sign does not change and the graph touches. Complex factors affect the polynomial's values but do not create real intercepts.

Worked example - annotate the reasoning

Problem. Analyze $P(x) = (x + 1)^2(x - 3)^3(x^2 + 4)$.

Model reasoning. Degree 7. Real zeros: -1 multiplicity 2 (touch) and 3 multiplicity 3 (cross with flattening). Nonreal zeros: $\pm 2i$. Positive leading coefficient and odd degree give left end down, right end up.

Check your understanding

Independent

No calculator

Analyze $(x - 2)^4(x + 5)(x^2 - 2x + 5)$.

Representation laboratory

Connect at least two representations

A real polynomial graph crosses at $x = -4$, touches at $x = 1$, and has no other real intercepts. Its degree is 7. Give two different possible factorizations consistent with the graph and explain how the unseen degrees are accounted for.

Challenge laboratory

Challenge - justify every claim

Construct a real-coefficient quartic whose zeros are $1 \pm i$ and $2 \pm 3i$, and whose value at 0 is 52. Give factored and expanded forms.

Error clinic and calculator strategy

Common errors to diagnose

- Counting only visible x -intercepts as all zeros.
- Forgetting the conjugate of a nonreal zero for real coefficients.
- Ignoring multiplicity when comparing zero count with degree.
- Saying the graph crosses the real axis at a complex zero.

Technology decision

A calculator can approximate complex zeros, but rounded roots should be checked by substitution or reconstructed factors. Numerical output can split repeated roots or hide exact forms, so use algebraic structure whenever coefficients factor cleanly.

AP-style synthesis**Multi-part reasoning task**

A monic degree-5 polynomial with real coefficients has zeros -2 , $1 + 2i$, and 3 with multiplicity 2.

A. List all five zeros counted with multiplicity.

B. Write the polynomial in factored form over the reals.

C. Expand the polynomial and state the only real x -intercepts with cross/touch behavior.

Exit ticket**Before you leave**

1. Why do nonreal zeros come in pairs for real coefficients?

2. What factor corresponds to $4 \pm 3i$?

3. How does even multiplicity affect a graph at a real zero?

Topic 1.6 - Polynomial Functions and End Behavior

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

End behavior describes the global direction of a polynomial far to the left and right. It is controlled by the first nonzero term of greatest degree, not by intercepts or middle coefficients. Careful parameter analysis is essential when a nominal leading term can cancel.

Learning targets

- Determine end behavior from degree parity and leading coefficient.
- Express end behavior with words, arrows, and limit notation and justify leading-term dominance.
- Analyze parameter families in which the degree and end behavior change through cancellation.

Language and structure

Term	Meaning in this topic
leading term	The nonzero term with greatest exponent.
leading coefficient	The coefficient of the leading term.
even-degree behavior	Both ends move in the same vertical direction.
odd-degree behavior	The two ends move in opposite directions.
leading-term dominance	Lower-degree terms become small relative to the leading term as $ x $ grows.

Launch - reason before calculating

Without graphing, determine the end behavior of $P(x) = -3x^5 + 2x^2 - 1$. Explain why the $2x^2$ term cannot make both ends rise.

Concept 1: Parity and leading coefficient

Core idea

Positive even leading terms rise on both ends; negative even leading terms fall on both ends. Positive odd leading terms fall left and rise right; negative odd leading terms rise left and fall right. Zeros and vertical shifts do not alter this global classification.

Worked example - annotate the reasoning

Problem. Classify $F(x) = 0.2x^6 - 4x^3 + 7$.

Model reasoning. The leading term $0.2x^6$ is positive even, so $F(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$.

Check your understanding

Independent

No calculator

Classify $G(x) = -5x^8 + 100x^7 - 1$.

Concept 2: Limit notation and dominance

Core idea

Factor the leading term to justify dominance: $P(x) = a_n x^n [1 + \frac{a_{n-1}}{a_n x} + \dots]$. The bracket tends to 1, so the sign and parity of $a_n x^n$ govern the limits. This is a proof, not merely a graph pattern.

Worked example - annotate the reasoning

Problem. Justify the right-end behavior of $H(x) = 4x^4 - 300x^3 + 2$.

Model reasoning. $H = 4x^4[1 - 75/x + 1/(2x^4)]$. The bracket tends to 1 as $x \rightarrow \infty$, so $H \rightarrow \infty$.

Check your understanding

Independent

No calculator

Write both end-behavior limits for $K(x) = 7 - 2x^3 + x$.

Concept 3: Parameter cancellation changes degree

Core idea

In a parameter family, determine the actual first nonzero coefficient before classifying end behavior. A special parameter value may cancel the expected leading term, exposing a lower-degree term with different parity or sign.

Worked example - annotate the reasoning

Problem. Classify $F_k(x) = (k - 2)x^5 + (3k + 1)x^4 - 7x + 9$ for $k = 2$ and $k \neq 2$.

Model reasoning. If $k = 2$, the x^5 term cancels and the leading term is $7x^4$, so both ends rise. If $k \neq 2$, degree 5 is odd; the right end follows the sign of $k - 2$ and the left end is opposite.

Check your understanding

Independent

No calculator

For $Q_a(x) = (a + 1)x^6 + (2 - a)x^5 + x^2$, find all a giving left end down and right end up.

Representation laboratory

Connect at least two representations

A graph rises on both far ends and has three visible real zeros. List all possible parities for its degree, state the sign of its leading coefficient, and explain why three intercepts do not determine the exact degree.

Challenge laboratory

Challenge - justify every claim

Find all parameter values a, b for which $P(x) = (a - 1)x^6 + (b + 2)x^5 - 4x^2 + 1$ has left end down and right end up. State whether the answer is unique.

Error clinic and calculator strategy

Common errors to diagnose

- Using the constant term or y -intercept to classify end behavior.
- Assuming the largest numerical coefficient creates the leading term.
- Ignoring cancellation at a special parameter value.
- Confusing end behavior with increasing/decreasing behavior on the whole real line.

Technology decision

A graphing window may not extend far enough for the leading term to dominate, especially when lower-degree coefficients are large. Use algebra first, then expand the window to confirm. Contextual models should be graphed only on their intended domain as well as globally for structural comparison.

AP-style synthesis**Multi-part reasoning task**

Let $P_a(x) = (a - 3)x^6 + (2a + 1)x^5 - 7x + 4$.

A. Classify end behavior for $a > 3$ and $a < 3$.

B. Determine the end behavior when $a = 3$.

C. Find every a for which the ends move in opposite directions and justify uniqueness.

Exit ticket**Before you leave**

1. State the four parity/sign end-behavior cases.

2. Why can a calculator window misrepresent end behavior?

3. What must be checked first in a parameter family?

Topic 1.7 - Rational Functions and End Behavior

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Rational functions often describe saturation, average cost, efficiency, and ratios. Their long-run behavior is not always a horizontal line: polynomial division can expose a slant or higher-degree asymptote, and the remainder describes the shrinking deviation from that model.

Learning targets

- Determine rational end behavior by comparing numerator and denominator degrees.
- Use polynomial division to identify slant or polynomial asymptotes.
- Analyze approach from above or below and interpret long-run behavior in context.

Language and structure

Term	Meaning in this topic
horizontal asymptote	A constant long-run model $y = L$.
slant asymptote	A linear long-run model found by division when degree difference is 1.
polynomial asymptote	The quotient polynomial when numerator degree exceeds denominator degree.
proper rational function	A rational expression with numerator degree below denominator degree.
remainder form	$R(x) = Q(x) + \frac{r(x)}{d(x)}$, making deviation from Q visible.

Launch - reason before calculating

Compare the long-run behavior of $A(x) = \frac{3x^2-5}{x^2+4}$ and $B(x) = \frac{3x^2-5}{x^3+4}$. Explain why the same numerator does not produce the same asymptote.

Concept 1: Degree comparison

Core idea

If numerator degree is less than denominator degree, the horizontal asymptote is $y = 0$. If degrees are equal, it is the ratio of leading coefficients. If numerator degree is greater, division is needed; there is no horizontal asymptote from a leading-coefficient ratio.

Worked example - annotate the reasoning

Problem. Find the asymptote of $R(x) = \frac{3x^2-5}{x^2+4}$ and determine whether the graph approaches from above or below.

Model reasoning. Asymptote $y = 3$. Since $R - 3 = \frac{-17}{x^2+4} < 0$, the graph approaches from below on both ends.

Check your understanding

Independent

No calculator

Find the horizontal asymptote of $S(x) = \frac{-4x^3+x}{2x^3-7}$.

Concept 2: Polynomial division and slant asymptotes

Core idea

When numerator degree is at least denominator degree, divide. The quotient is the polynomial asymptote because the proper remainder term tends to zero when its numerator has lower degree than the denominator.

Worked example - annotate the reasoning

Problem. Rewrite $\frac{2x^2+3x-1}{x-2}$ in quotient-remainder form.

Model reasoning. Division gives $2x + 7 + \frac{13}{x-2}$, so the slant asymptote is $y = 2x + 7$.

Check your understanding

Independent

No calculator

Find the polynomial asymptote of $\frac{x^3-2x+5}{x^2+1}$.

Concept 3: Approach and contextual meaning

Core idea

The sign of $R(x) - Q(x)$ shows whether the graph lies above or below its asymptote. In context, the asymptote may represent limiting cost, concentration, capacity, or a trend line. A horizontal asymptote can be crossed; it is not a domain restriction.

Worked example - annotate the reasoning

Problem. For $F(x) = 5 + \frac{2x}{x^2+1}$, describe approach to $y = 5$ on both ends.

Model reasoning. For large positive x , the correction is positive, so approach from above. For large negative x , it is negative, so approach from below.

Check your understanding

Independent

No calculator

For $G(x) = 8 - \frac{3}{x^2+4}$, describe the approach.

Representation laboratory

Connect at least two representations

A graph follows the line $y = -2x + 3$ more closely as $|x|$ grows. Write two different rational functions with that slant asymptote, one above the line for large positive x and one below it.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with horizontal asymptote $y = -2$, approached from above as $x \rightarrow \infty$ and from below as $x \rightarrow -\infty$, with no real vertical asymptotes.

Error clinic and calculator strategy

Common errors to diagnose

- Using leading-coefficient ratio when degrees are unequal.
- Stopping division before the remainder degree is below the divisor degree.
- Treating an asymptote as a line the graph can never cross.
- Ignoring the sign of the remainder when describing approach.

Technology decision

Graph the rational function together with its asymptote and graph their difference. A wide window can show long-run agreement, while a difference table reveals approach direction. Algebraic division remains the most reliable way to identify the exact asymptote.

AP-style synthesis**Multi-part reasoning task**

Let $R(x) = \frac{2x^3 - x^2 + 5}{x^2 - 4}$.

- A. Use division to write R in quotient-remainder form.

- B. State the polynomial asymptote and excluded inputs.

- C. Determine whether R is above or below its asymptote for sufficiently large positive x .

Exit ticket**Before you leave**

1. State the three degree-comparison cases.

2. What part of division gives the polynomial asymptote?

3. Can a rational graph cross a horizontal asymptote?

Topic 1.8 - Rational Functions and Zeros

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A numerator value of zero creates a rational-function zero only when the denominator is nonzero in the original function. Simplification may reveal true zeros but can also hide excluded inputs. Multiplicity controls whether the graph crosses or touches the axis.

Learning targets

- Determine rational-function zeros while preserving original domain restrictions.
- Use multiplicity and sign charts to describe behavior near zeros.
- Construct rational functions from specified zeros, holes, asymptotes, and scale conditions.

Language and structure

Term	Meaning in this topic
rational zero	An allowed input where the numerator is zero and denominator is nonzero.
excluded value	An input that makes the original denominator zero.
canceled factor	A common factor whose input remains excluded after simplification.
zero multiplicity	The exponent of an uncanceled numerator factor.
sign chart	An interval analysis of whether function values are positive or negative.

Launch - reason before calculating

A student says $x = 3$ is a zero of $R(x) = \frac{(x-3)(x+1)}{(x-3)(x-5)}$ because it makes the numerator zero. Evaluate the claim and identify the actual zero.

Concept 1: Zeros after simplification, restrictions retained

Core idea

Factor numerator and denominator completely. Cancel common factors to reveal the simplified rule, but list original exclusions first. A zero must come from an uncanceled numerator factor and must remain in the domain.

Worked example - annotate the reasoning

Problem. Find all zeros and exclusions of $F(x) = \frac{(x-2)(x+4)}{(x+4)(x-7)}$.

Model reasoning. Exclusions are $x = -4, 7$. Simplified rule $(x-2)/(x-7)$ has zero $x = 2$. The canceled value -4 is a hole.

Check your understanding

Independent

No calculator

Find zeros and exclusions of $G(x) = \frac{x^2-9}{x^2-x-6}$.

Concept 2: Multiplicity and sign at zeros

Core idea

At an odd-multiplicity zero, the rational function changes sign if no other factor changes there; at an even-multiplicity zero it keeps sign and touches the axis. A sign chart should include zeros and vertical asymptotes as interval boundaries, while holes do not necessarily change sign.

Worked example - annotate the reasoning

Problem. Analyze zeros of $S(x) = \frac{(x+2)^2(x-1)}{(x-4)^3}$.

Model reasoning. Zeros are -2 multiplicity 2 (touch/no sign change) and 1 multiplicity 1 (cross/sign change). $x = 4$ is a vertical asymptote, not a zero.

Check your understanding

Independent

No calculator

Describe zero behavior of $T(x) = \frac{(x-5)^3}{(x^2+1)(x+2)^2}$.

Concept 3: Constructing rational functions

Core idea

Use numerator factors for zeros, denominator factors for vertical asymptotes, common factors for holes, and positive quadratics when degree is needed without adding real features. A given point determines the scale factor.

Worked example - annotate the reasoning

Problem. Construct a function with zero 2 simple, zero -3 double, vertical asymptotes 1, -4 , and value $R(0) = 9$.

Model reasoning. Let $R = k(x - 2)(x + 3)^2 / [(x - 1)(x + 4)]$. At 0, the factor ratio is $18/4 = 4.5$, so $k = 2$.

Check your understanding

Independent

No calculator

Construct a rational function with zero 1 double, vertical asymptote -2 , and no other real zeros or asymptotes.

Representation laboratory

Connect at least two representations

A graph crosses the x -axis at -1 , touches at 3, has a hole at 2, and has a vertical asymptote at -4 . Write one factored rational function consistent with these features and state every original exclusion.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with horizontal asymptote $y = 2$, zero -2 of multiplicity 2, zero 3 simple, hole at 1, and vertical asymptote at -4 , with no other real features.

Error clinic and calculator strategy

Common errors to diagnose

- Calling every numerator root a zero without checking the original denominator.
- Forgetting that canceled inputs remain excluded.
- Using denominator roots as x -intercepts.
- Ignoring zero multiplicity in sign and graph behavior.

Technology decision

A graphing utility may display a canceled hole poorly or fill it visually. Factor algebraically, inspect a table near the excluded input, and graph the original and simplified rules together. Zeros should be verified by substitution into the original domain.

AP-style synthesis**Multi-part reasoning task**

Let $F(x) = \frac{(x-4)^2(x+1)}{(x-2)(x+1)(x^2+1)}$.

A. State the domain exclusions and simplified rule.

B. Identify all zeros with multiplicity and graph behavior.

C. Determine the sign of F on each interval formed by its real zero and vertical asymptote.

Exit ticket**Before you leave**

1. What two conditions must an input satisfy to be a rational zero?

2. What does even zero multiplicity do to sign?

3. Why must restrictions be recorded before cancellation?

Topic 1.9 - Rational Functions and Vertical Asymptotes

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A vertical asymptote describes unbounded output near a finite excluded input. Identifying the asymptote is only the first step: strong analysis explains each one-sided limit by simplifying, checking multiplicity, and tracking signs rather than relying on a sketch.

Learning targets

- Identify vertical asymptotes after cancellation and distinguish them from holes.
- Determine one-sided infinite behavior using sign and multiplicity.
- Construct and interpret rational functions with specified asymptotic behavior.

Language and structure

Term	Meaning in this topic
vertical asymptote	An excluded input near which output magnitude grows without bound.
one-sided limit	Behavior as x approaches from values less than or greater than the input.
even denominator multiplicity	The denominator factor keeps the same sign on both sides.
odd denominator multiplicity	The denominator factor changes sign across the asymptote.
divergence	Unbounded growth toward $+\infty$ or $-\infty$.

Launch - reason before calculating

Compare $R(x) = \frac{x+1}{x-2}$ and $S(x) = \frac{x-3}{(x+1)^2}$ near their vertical asymptotes. Why do the one-sided directions differ?

Concept 1: Simplify before identifying asymptotes

Core idea

Factor and cancel common factors while retaining restrictions. Uncanceled real denominator zeros are vertical asymptotes; canceled denominator zeros are holes. Positive quadratics may add no real asymptotes.

Worked example - annotate the reasoning

Problem. Classify excluded inputs of $F = \frac{(x-1)(x+2)}{(x-1)(x-4)}$.

Model reasoning. $x = 1$ is a hole after cancellation; $x = 4$ remains an uncanceled denominator zero and is a vertical asymptote.

Check your understanding

Independent

No calculator

Classify $G = \frac{x^2-9}{(x-3)^2(x+1)}$.

Concept 2: One-sided signs and multiplicity

Core idea

Near $x = a$, freeze factors that stay nonzero at their signs and focus on $(x - a)^m$. If m is odd, denominator sign changes; if even, it does not. The nearby numerator sign determines whether the unbounded values are positive or negative.

Worked example - annotate the reasoning

Problem. Find one-sided behavior of $R = \frac{x+1}{x-2}$ at 2.

Model reasoning. Numerator is near $3 > 0$. $x - 2$ is negative left and positive right, so $R \rightarrow -\infty$ from the left and $+\infty$ from the right.

Check your understanding

Independent

No calculator

Find one-sided behavior of $S = \frac{x-3}{(x+1)^2}$ at -1 .

Concept 3: Multiple asymptotes and construction

Core idea

Analyze each asymptote locally; other factors contribute fixed signs. To construct a requested pattern, choose denominator multiplicities and a numerator whose sign at each asymptote creates the required directions.

Worked example - annotate the reasoning

Problem. Analyze $T = \frac{x-1}{(x+2)^2(x-3)}$ at -2 and 3 .

Model reasoning. Near -2 , numerator and $x - 3$ are both negative, and the square is positive, so both sides $+\infty$. Near 3 , numerator and square are positive, while $x - 3$ changes sign: left $-\infty$, right $+\infty$.

Check your understanding

Independent

No calculator

Construct a function with vertical asymptote 0 and both sides approaching $-\infty$.

Representation laboratory

Connect at least two representations

A graph has vertical asymptote $x = 1$ with left $+\infty$ and right $-\infty$, and asymptote $x = -3$ with both sides $+\infty$. Give one factored rational function that matches.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with zero 1, vertical asymptote -2 approached by $+\infty$ from both sides, and vertical asymptote 3 approached by $-\infty$ from the left and $+\infty$ from the right.

Error clinic and calculator strategy

Common errors to diagnose

- Calling a canceled denominator zero a vertical asymptote.
- Stating $\pm\infty$ without separate one-sided analysis.
- Using denominator multiplicity alone while ignoring numerator and other-factor signs.
- Treating infinity as an output value attained by the function.

Technology decision

Use a table approaching from both sides with successively smaller distances and graph separate branches. Auto-scaling can hide direction or make a hole look like an asymptote. Exact sign analysis should lead; technology should confirm.

AP-style synthesis**Multi-part reasoning task**

Let $R(x) = \frac{(x+4)(x-1)}{(x+2)^2(x-3)}$.

- A. Identify zeros and vertical asymptotes.

- B. Determine both one-sided limits at $x = -2$.

- C. Determine both one-sided limits at $x = 3$.

Exit ticket**Before you leave**

1. What distinguishes a hole from a vertical asymptote?

2. How does even denominator multiplicity affect side signs?

3. Why are one-sided limits necessary?

Topic 1.10 - Rational Functions and Holes

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A hole is not simply a missing dot on a graph; it records an input excluded by the original rule even though a simplified expression has a finite nearby value. This distinction matters when simplifying, graphing, repairing a function, and interpreting equivalent expressions.

Learning targets

- Identify holes from canceled factors and determine exact coordinates.
- Distinguish holes from zeros and vertical asymptotes while preserving the original domain.
- Define repaired piecewise functions and construct rational functions with specified holes.

Language and structure

Term	Meaning in this topic
hole	A removable discontinuity caused by a canceled denominator factor.
removable discontinuity	An excluded input with a finite limiting value.
hole coordinate	$(a, g(a))$, where a is excluded and g is the simplified rule.
continuous repair	Defining the missing output to equal the simplified value.
equivalent on a domain	Two rules agree for all inputs in the stated common domain.

Launch - reason before calculating

Analyze $R(x) = \frac{x^2-9}{x^2-x-6}$. Identify every excluded input and determine which is a hole and which is a vertical asymptote.

Concept 1: Finding a hole and its coordinate

Core idea

Record original restrictions, cancel common factors, then substitute the canceled input into the simplified rule. The resulting finite value is the hole's y -coordinate, not a value of the original function.

Worked example - annotate the reasoning

Problem. Find the hole of $R = \frac{x^2-9}{x^2-x-6}$.

Model reasoning. Simplified $(x+3)/(x+2)$ with $x \neq 3, -2$. At $x = 3$, the simplified value is $6/5$, so the hole is $(3, 6/5)$.

Check your understanding

Independent

No calculator

Find the hole of $F = \frac{(x-1)(x+4)}{(x-1)(x-2)}$.

Concept 2: Holes, zeros, and remaining factors

Core idea

After cancellation, a canceled input can lie on the x -axis of the simplified graph, but it remains a hole rather than a zero. If part of a repeated denominator factor remains after cancellation, the input is still a vertical asymptote, not a hole.

Worked example - annotate the reasoning

Problem. Analyze $G = \frac{(x-1)^2(x+4)}{(x-1)(x-2)}$.

Model reasoning. Simplified $(x-1)(x+4)/(x-2)$. The canceled input 1 creates a hole at $(1, 0)$; it is not a zero of the original function. $x = 2$ is a vertical asymptote.

Check your understanding

Independent

No calculator

Analyze $H = \frac{(x+3)(x-2)}{(x+3)^2}$.

Concept 3: Repair and construction

Core idea

A hole at $x = a$ is repaired by defining the missing value to equal the simplified expression at a . To construct a hole, multiply numerator and denominator by $(x - a)$; choose the simplified rule first so that its value at a is the desired height.

Worked example - annotate the reasoning

Problem. Repair $R = \frac{x^2-9}{x^2-x-6}$ at $x = 3$.

Model reasoning. Define $F(x) = R(x)$ for $x \neq 3, -2$ and $F(3) = 6/5$. The vertical asymptote at -2 cannot be repaired by assigning one finite value.

Check your understanding

Independent

No calculator

Construct a rational function with hole $(2, 5)$, vertical asymptote -1 , and zero 4 .

Representation laboratory

Connect at least two representations

A graph looks like $y = (x + 1)/(x - 3)$ but has open circles at $x = 0$ and $x = 2$. Write one original rational expression that creates exactly those holes and state their coordinates.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with holes at $(1, 2/5)$ and $(2, 1/2)$, vertical asymptote -4 , and no zeros other than the zero of its simplified numerator.

Error clinic and calculator strategy

Common errors to diagnose

- Substituting into the original $0/0$ form to obtain a hole coordinate.
- Calling a canceled input a zero because the simplified numerator also vanishes there.
- Forgetting noncanceled denominator factors after partial cancellation.
- Claiming the original and simplified rules have identical domains.

Technology decision

Graphing software may not display a tiny hole. Compare original and simplified tables, inspect the original domain, and use an open-circle annotation. Exact factoring is more reliable than visual detection.

AP-style synthesis**Multi-part reasoning task**

Let $R(x) = \frac{(x-1)(x+2)^2}{(x-1)(x+2)(x-3)}$.

A. State all original exclusions and simplify.

B. Identify every hole and its coordinate and every vertical asymptote.

C. Define a piecewise repair at all removable discontinuities.

Exit ticket**Before you leave**

1. How is a hole's y -coordinate found?

2. Can a hole lie on the x -axis without being a zero?

3. What value repairs a removable discontinuity?

Topic 1.11 - Equivalent Representations of Polynomial and Rational Expressions

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

The same polynomial or rational expression can be written in several useful forms. Expanded form emphasizes degree and leading behavior; factored form reveals zeros and multiplicities; quotient-remainder form reveals division structure and long-run behavior. Algebraic equivalence must be justified, and canceled factors never erase original domain restrictions.

Learning targets

- Rewrite polynomial expressions in factored, expanded, and quotient-remainder forms and explain what each form reveals.
- Use polynomial division and the Remainder and Factor Theorems to make and justify conclusions.
- Rewrite rational expressions while preserving exclusions, holes, zeros, and asymptotes.

Language and structure

Term	Meaning in this topic
equivalent expressions	Expressions with the same value at every input common to their domains.
factor form	A product form that makes zeros and multiplicities visible.
quotient-remainder form	$P(x) = D(x)Q(x) + R(x)$, where the degree of R is less than the degree of D .
Remainder Theorem	The remainder when $P(x)$ is divided by $x - c$ equals $P(c)$.
domain restriction	An input excluded by the original expression, even if a factor later cancels.

Launch - reason before calculating

The expressions $x^3 - x^2 - 8x + 12$ and $(x - 2)^2(x + 3)$ look different. Without graphing, list one feature that is easier to read from each form and explain why both forms should be retained in a complete analysis.

Concept 1: Selecting a form that reveals structure

Core idea

Factoring and expanding are reversible when performed correctly. Use factored form for zeros, multiplicity, sign, and cancellation; expanded form for degree, leading coefficient, coefficient comparison, and some regression models. A complete solution often moves between forms rather than treating one as universally best.

Worked example - annotate the reasoning

Problem. Rewrite $P(x) = x^3 - 4x^2 - x + 4$ in factored form and state what each representation reveals.

Model reasoning. Group terms: $x^2(x - 4) - 1(x - 4) = (x - 4)(x^2 - 1) = (x - 4)(x - 1)(x + 1)$. Expanded form reveals degree 3 and positive leading coefficient. Factored form reveals simple zeros $-1, 1, 4$.

Check your understanding

Independent

No calculator

Factor $x^4 - 5x^2 + 4$ completely over the reals and identify all real zeros with multiplicity.

Concept 2: Division, remainders, and factor tests

Core idea

Polynomial division produces an identity valid for every input: $P = DQ + R$. For a linear divisor $x - c$, the remainder is the number $P(c)$. Thus $x - c$ is a factor exactly when $P(c) = 0$. Synthetic division is efficient, but the quotient-remainder identity should still be stated.

Worked example - annotate the reasoning

Problem. Divide $2x^3 - 3x^2 + 4x - 5$ by $x - 2$ and verify the remainder independently.

Model reasoning. Synthetic division with 2 gives quotient $2x^2 + x + 6$ and remainder 7. Therefore $P = (x - 2)(2x^2 + x + 6) + 7$. Also $P(2) = 16 - 12 + 8 - 5 = 7$.

Check your understanding

Independent

No calculator

Divide $x^4 - 3x^3 + 2x^2 + 5x - 7$ by $x + 1$ and interpret the remainder.

Concept 3: Rational equivalence and preserved restrictions

Core idea

Canceling a common factor changes the written expression but not the original domain. The simplified expression describes the graph except at canceled inputs. A canceled denominator factor creates a hole; an uncanceled denominator factor can create a vertical asymptote. State exclusions before canceling.

Worked example - annotate the reasoning

Problem. Simplify $\frac{x^2-9}{x^2-x-6}$ and identify every discontinuity.

Model reasoning. The original denominator factors as $(x-3)(x+2)$, so $x=3, -2$ are excluded. Canceling $x-3$ gives $(x+3)/(x+2)$. There is a hole at $(3, 6/5)$ and a vertical asymptote at $x=-2$.

Check your understanding

Independent

No calculator

Simplify $\frac{x^2-4x+4}{x^2-4}$ and identify the zero, hole, and vertical asymptote.

Representation laboratory

Connect at least two representations

For $P(x) = (x-2)^2(x+3) = x^3 - x^2 - 8x + 12$, create a two-column feature inventory: features visible immediately in factored form and features visible immediately in expanded form. Then identify one feature requiring additional reasoning in either form.

Challenge laboratory

Challenge - justify every claim

Expand $(2x - 3)^4$ using structure rather than repeated multiplication. Then explain how the coefficient pattern is connected to Pascal's triangle and verify the constant and leading terms without completing the full expansion again.

Error clinic and calculator strategy

Common errors to diagnose

- Canceling terms across addition, such as changing $(x + 3)/x$ into 3.
- Declaring two rational functions identical without noting an excluded input in one domain.
- Using synthetic division with the wrong sign for a divisor such as $x + 4$.
- Reporting a factor from an approximate zero without checking the remainder.

Technology decision

A computer algebra system can factor, expand, and divide, but it may suppress original restrictions or return a form that hides the requested feature. Record the original domain first, and verify a claimed factor with substitution or multiplication.

AP-style synthesis**Multi-part reasoning task**

Let $P(x) = x^4 - 5x^3 + 2x^2 + 20x - 24$.

- A. Verify that $P(x) = (x - 3)(x - 2)^2(x + 2)$ by a reversible algebraic method.

- B. Use the most informative form to identify all zeros and graph behavior at each zero.

- C. Rewrite $\frac{P(x)}{(x-2)(x+2)}$ in simplest form while preserving all original restrictions, and identify every hole.

Exit ticket**Before you leave**

1. What does the remainder on division by $x - c$ equal?

2. Why can canceling a factor create a hole rather than remove a restriction?

3. Which form is usually best for identifying multiplicity?

Topic 1.12 - Transformations of Functions

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Transformation notation compresses a great deal of geometric information. Strong reasoning distinguishes horizontal and vertical actions, explains the apparent reversal inside the input, maps points exactly, and tracks features such as domain, range, asymptotes, and excluded inputs rather than relying only on a memorized list of moves.

Learning targets

- Interpret and construct transformations in the form $g(x) = af(b(x - h)) + k$.
- Map points and key features between parent and transformed functions.
- Determine transformed domains, ranges, intercepts, asymptotes, and holes with justification.

Language and structure

Term	Meaning in this topic
horizontal translation	The shift controlled by h inside $f(b(x - h))$.
vertical translation	The shift controlled by k outside the function.
horizontal scale	Input distances are multiplied by $1/ b $.
vertical scale	Output distances are multiplied by $ a $.
point mapping	If (u, v) lies on f , then $(h + u/b, k + av)$ lies on g .

Launch - reason before calculating

A point $(-2, 5)$ lies on $y = f(x)$. Without graphing the full function, determine the corresponding point on $g(x) = -3f(2(x - 1)) + 4$ and explain the role of every parameter.

Concept 1: Reading and mapping a general transformation

Core idea

For $g(x) = af(b(x - h)) + k$, solve $b(x - h) = u$ to map an original input u . Thus (u, v) maps to $(h + u/b, k + av)$. A negative b reflects horizontally, a negative a reflects vertically, and the horizontal scale is reciprocal.

Worked example - annotate the reasoning

Problem. If $(6, -1)$ lies on f , find its image on $h(x) = 2f((x + 4)/3) - 5$.

Model reasoning. Set $(x + 4)/3 = 6$, so $x = 14$. The output becomes $2(-1) - 5 = -7$. The image is $(14, -7)$.

Check your understanding

Independent

No calculator

If $(-3, 4)$ lies on f , find its image on $g(x) = -2f(-4(x - 1)) + 7$.

Concept 2: Transforming parent graphs, domains, and ranges

Core idea

Transform key features, not just isolated points. Vertical shifts change range and horizontal asymptotes; horizontal shifts change domain landmarks and vertical asymptotes. Reflections and stretches change orientation and scale. For a quadratic in vertex form, the transformed vertex and axis are immediately visible.

Worked example - annotate the reasoning

Problem. Analyze $g(x) = -2(x + 3)^2 + 5$.

Model reasoning. From $y = x^2$, shift left 3, stretch vertically by 2, reflect across the x -axis, and shift up 5. Vertex $(-3, 5)$, axis $x = -3$, domain all real numbers, range $y \leq 5$.

Check your understanding

Independent

No calculator

Analyze $q(x) = 3|x - 2| - 4$, including vertex, opening, domain, and range.

Concept 3: Asymptotes, holes, and transformed restrictions

Core idea

For $r(x) = a/(x - h) + k$, the vertical asymptote is $x = h$ and horizontal asymptote is $y = k$. A transformed excluded input is found by solving the inside expression equal to the original excluded input. A transformed hole's coordinate follows the same point mapping as any other feature.

Worked example - annotate the reasoning

Problem. Analyze $r(x) = -\frac{2}{x-4} + 3$.

Model reasoning. Vertical asymptote $x = 4$, horizontal asymptote $y = 3$, domain $x \neq 4$, range $y \neq 3$. The negative scale reflects the reciprocal branches; for $x > 4$, values lie below 3.

Check your understanding

Independent

No calculator

For $s(x) = \frac{3}{2(x+1)} - 5$, identify asymptotes, domain, and range.

Representation laboratory

Connect at least two representations

The table contains points of f : $(-4, 2), (-2, -1), (0, 3), (6, 5)$. Construct the corresponding table for $g(x) = 2f((x - 1)/3) - 4$ and describe how input spacing and output spacing change.

Challenge laboratory

Challenge - justify every claim

A transformed function $g(x) = af(b(x - h)) + k$ maps the points $(0, 2)$ and $(4, -1)$ on f to $(5, 7)$ and $(3, 1)$ on g . Determine a, b, h, k if $a > 0$ and explain whether the parameters are unique from the stated information.

Error clinic and calculator strategy

Common errors to diagnose

- Treating $f(2x)$ as a horizontal stretch by 2 instead of a compression by $1/2$.
- Shifting a graph before accounting for the full grouped input $b(x - h)$.
- Transforming points but forgetting to transform domain endpoints or excluded inputs.
- Moving a horizontal asymptote horizontally when a horizontal shift does not change its y -value.

Technology decision

Graphing technology is useful for checking a transformation, but a window can hide a vertex or asymptote. Predict mapped features first, then choose a window containing those features and verify that the graph agrees.

AP-style synthesis**Multi-part reasoning task**

The points $(-4, 3)$, $(-1, -2)$, $(2, 5)$ lie on f . Define $g(x) = -2f(3(x - 1)) + 4$.

A. Map all three points to g .

B. If the domain of f is $[-4, 2]$, determine the domain of g .

C. If the range of f is $[-2, 5]$, determine the range of g and explain the reversal of endpoints.

Exit ticket**Before you leave**

1. What is the point map for $g(x) = af(b(x - h)) + k$?

2. Why does $f(4x)$ compress horizontally?

3. What are the asymptotes of $a/(x - h) + k$?

Topic 1.13 - Function Model Selection and Assumption Articulation

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Many models can fit a small data set. A defensible choice combines patterns in differences or ratios, residual behavior, contextual mechanisms, and a realistic domain. Equally important, assumptions and limitations must be stated so that a mathematical fit is not mistaken for certainty outside the conditions under which the data were collected.

Learning targets

- Use differences, ratios, rates, and graphical shape to compare candidate model families.
- Compute and interpret residuals and use residual patterns to evaluate fit.
- State assumptions, contextual domains, interpolation limits, and extrapolation risks in precise language.

Language and structure

Term	Meaning in this topic
residual	Observed value minus model-predicted value.
interpolation	Prediction within the span of observed inputs.
extrapolation	Prediction beyond the observed input range.
assumption	A condition treated as true so the model can be used.
model adequacy	Whether a model is accurate and appropriate enough for its intended decision.

Launch - reason before calculating

Two models have nearly identical coefficients of determination for a data set. Explain why that statistic alone cannot determine which model should be used for a ten-year prediction.

Concept 1: Structural evidence from differences and ratios

Core idea

Approximately constant first differences support a linear model; approximately constant second differences at equally spaced inputs support a quadratic model; approximately constant ratios support an exponential model. These patterns are evidence, not automatic proof, especially with noisy or sparse data.

Worked example - annotate the reasoning

Problem. For $x = 0, 1, 2, 3, 4$, outputs are 3, 7, 15, 27, 43. Select a model family and construct an exact model.

Model reasoning. First differences are 4, 8, 12, 16 and second differences are constant at 4, so use a quadratic with $2a = 4$, giving $a = 2$. Solving gives $y = 2x^2 + 2x + 3$.

Check your understanding

Independent

No calculator

Outputs at equally spaced inputs are 5, 7.5, 11.25, 16.875. Select a model family and write a model beginning at input 0.

Concept 2: Residuals as a diagnostic, not decoration

Core idea

A residual is $y - \hat{y}$. Positive residuals mean the model underpredicted; negative residuals mean it overpredicted. A suitable model tends to produce small residuals scattered without a systematic curve, trend, or change in spread. A high R^2 can coexist with a problematic residual pattern.

Worked example - annotate the reasoning

Problem. Model A has residuals $-4, 1, 5, 2, -4$; Model B has $0.2, -0.3, 0.1, 0.2, -0.2$. Compare the models.

Model reasoning. Model B has much smaller residuals and no strong visible pattern. Model A shows large edge errors and a middle bulge, suggesting missing curvature. Model B is better on this evidence.

Check your understanding

Independent

No calculator

An observed value is 42.7 and the model predicts 45.1. Find and interpret the residual.

Concept 3: Context, assumptions, and responsible prediction

Core idea

A model is used only on a defensible domain and under stated assumptions. Identify what is held stable, how the data were measured, whether inputs are continuous or discrete, and whether outside forces may change. Interpolation is usually less risky than extrapolation, but neither is guaranteed.

Worked example - annotate the reasoning

Problem. A medicine-decay model was fitted for the first 12 hours in healthy adults. A student predicts concentration after 10 days. Evaluate the decision.

Model reasoning. The prediction is extreme extrapolation and ignores biological changes, measurement limits, dosing, and population differences. The fitted equation may be mathematically evaluable but not scientifically defensible there.

Check your understanding

Independent

No calculator

A quadratic revenue model is based on prices from 20 to 50 riyals. State two assumptions needed to use it at 45 riyals and one reason not to use it at 500 riyals.

Representation laboratory

Connect at least two representations

For inputs 0, 1, 2, 3, 4, a data set has outputs 6, 9, 13.5, 20.25, 30.375. Represent the pattern using a table of ratios, an equation, and a verbal interpretation of the parameter 1.5.

Challenge laboratory

Challenge - justify every claim

On $0 \leq x \leq 4$, the models $L(x) = 3x + 2$ and $Q(x) = 0.05x^2 + 2.8x + 2.1$ produce similar values. Compare their predictions at $x = 40$ and explain what the comparison teaches about extrapolation.

Error clinic and calculator strategy

Common errors to diagnose

- Choosing the model with the largest R^2 while ignoring a patterned residual plot.
- Calling a relationship causal because a regression fits well.
- Using a continuous domain for a count of objects without explanation.
- Reporting an extrapolated value without discussing whether assumptions remain plausible.

Technology decision

Technology can compute regressions and residual plots quickly. Enter data carefully, inspect axes and residual scale, and compare models using both numerical and contextual evidence. Save enough precision for calculations, then round only the final reported result appropriately.

AP-style synthesis**Multi-part reasoning task**

A data set has inputs 0, 1, 2, 3, 4, 5 and observed outputs 4.1, 7.0, 14.2, 24.8, 40.1, 59.0. Quadratic predictions are 4, 7, 14, 25, 40, 59; exponential predictions are 4.8, 7.5, 11.7, 18.3, 28.6, 44.7.

A. Compute the residual at input 3 for each model and interpret its sign.

B. Select the more defensible model using the full residual evidence.

C. State one assumption and one limitation if the selected model is used to predict at input 8.

Exit ticket**Before you leave**

1. What does a negative residual mean?

2. Which difference pattern suggests a quadratic model?

3. Why is extrapolation usually riskier than interpolation?

Topic 1.14 - Function Model Construction and Application

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Constructing a model is not only fitting an equation. It requires choosing a structure, using conditions to determine parameters, checking that the model satisfies those conditions, applying it to answer a question, interpreting rates and values with units, and evaluating whether the result is meaningful in context.

Learning targets

- Construct linear, quadratic, polynomial, rational, and piecewise models from stated conditions or data.
- Use models to solve equations, compare rates, and interpret parameters and outputs in context.
- Validate a model against conditions, domain, residual evidence, assumptions, and practical constraints.

Language and structure

Term	Meaning in this topic
parameter	A constant in a model whose value controls a structural feature.
calibration	Determining parameter values from data or conditions.
validation	Checking a model against observations or requirements not merely used to create it.
piecewise model	A model using different rules on different parts of the domain.
contextual domain	The input values for which the model and its assumptions are meaningful.

Launch - reason before calculating

A parabola has vertex $(3, 20)$ and passes through $(0, 2)$. Construct its equation and explain why the information determines a unique quadratic in vertex form.

Concept 1: Constructing from geometric and algebraic conditions

Core idea

Choose a form aligned to the information: slope-intercept for slope and intercept, vertex form for a quadratic vertex, factored form for zeros, and transformed rational form for asymptotes. Substitute remaining points to determine scale, then verify every condition.

Worked example - annotate the reasoning

Problem. Construct the quadratic with vertex $(3, 20)$ through $(0, 2)$.

Model reasoning. $y = a(x - 3)^2 + 20$ and $2 = 9a + 20$, so $a = -2$. The model is $y = -2(x - 3)^2 + 20$.

Check your understanding

Independent

No calculator

Construct the quadratic with vertex $(-2, 5)$ through $(0, 13)$.

Concept 2: Calibrating a rational model from asymptotes and values

Core idea

A quantity approaching a long-run level L can often be modeled by $M(t) = L + A/(t + b)$. The horizontal asymptote fixes L ; two observed values can determine A and b . Restrictions and the contextual domain must be stated separately.

Worked example - annotate the reasoning

Problem. A model approaches 80, with $M(0) = 180$ and $M(5) = 130$. Find $M(t) = 80 + A/(t + b)$.

Model reasoning. $A/b = 100$, so $A = 100b$. Also $A/(b + 5) = 50$, so $A = 50b + 250$. Hence $b = 5$, $A = 500$, and $M = 80 + 500/(t + 5)$.

Check your understanding

Independent

No calculator

A model approaches 20, with $N(0) = 70$ and $N(5) = 45$. Determine $N(t) = 20 + A/(t + b)$.

Concept 3: Piecewise construction, application, and validation

Core idea

Piecewise models encode changes in rules. Define each interval clearly, check endpoints and continuity when appropriate, and use the correct branch. After solving, interpret units and test whether the answer lies in the branch and contextual domain.

Worked example - annotate the reasoning

Problem. A parking garage charges 8 riyals for up to 3 hours and then 2.5 riyals per additional hour. Construct a continuous model for $0 \leq h \leq 12$.

Model reasoning. $C(h) = 8$ for $0 \leq h \leq 3$ and $C(h) = 8 + 2.5(h - 3)$ for $3 < h \leq 12$. Both branches agree at $h = 3$, so the model is continuous there.

Check your understanding

Independent

No calculator

Use the model to find the charge for 7.5 hours and solve for the time producing a charge of 25.50.

Representation laboratory

Connect at least two representations

A bridge arch passes through $(0, 0)$, $(5, 10)$, $(10, 0)$. Construct the model using both factored and vertex forms, and explain what each form makes visible.

Challenge laboratory

Challenge - justify every claim

Construct a least-degree real polynomial with zeros -2 simple and 3 double, with $P(0) = 36$ and right end down. State the degree, leading coefficient, and a reasonable sign chart.

Error clinic and calculator strategy

Common errors to diagnose

- Using a convenient form that does not encode the supplied features efficiently.
- Solving for a parameter but failing to verify all original conditions.
- Reporting an algebraic solution that lies outside the branch or contextual domain.
- Treating a fitted equation as exact law without checking residuals or assumptions.

Technology decision

Regression can determine parameters from many data points, but report the model family, window, residual evidence, and rounding policy. Keep full calculator precision internally, validate against data, and interpret only to a precision supported by the measurements.

AP-style synthesis**Multi-part reasoning task**

A pedestrian arch touches the ground at horizontal positions $x = 0$ and $x = 10$ and reaches height 10 at $x = 5$.

A. Construct a quadratic height model in factored form.

B. Rewrite the model in vertex form and identify the maximum.

C. Find the average rate of change from $x = 2$ to $x = 8$ and explain why the value can be zero although the arch is not horizontal throughout.

Exit ticket**Before you leave**

1. Which quadratic form is best when zeros are given?

2. What must be checked after solving with a piecewise model?

3. Why should regression coefficients not be rounded too early?
